Financial Processes - Homework 5 calculations

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Introduction

Since previously I haven't learned stochastic calculus, I used AI and mainly Wikipedia to (double) check the more important formulas, but I spent a lot of time trying to understand everything in the assignment. I wrote down every small step by pen and paper, here I skip some of them, but keep the main ones. I met some new theorems, so I already gained some knowledge from them. I hope my solution will be correct, and the explanations in this file are understandable. I think it is easier to understand what I did in the code if the one checking it reads this file next to it.

Part a)

Question. *What is the formula for $V(0)$?*

Consider $\max\{S_1(T), S_2(T)\} = S_2(T) + (S_1(T) - S_2(T))^+$, where $a^+ = \max(a, 0)$. If I type in Google: "price of $\max\{0, S_1(T) - S_2(T)\}$", it tells me the Margrabe-formula. It can handle correlation, but since S_1 and S_2 are independent, by the exercise $\sigma_1 = \sigma_2 = \sigma$, we can say the following:

$$\sigma_{new} = \sqrt{\sigma_1^2 + \sigma_2^2} = \sqrt{2\sigma^2} = \sqrt{2}\sigma$$

If Φ is the CDF if the standard normal distribution, by the exercise $S_1(0) = S_2(0) = 100$, the Margrabe-formula yields $\widetilde{V}(0) = S_1(0)\Phi(d_1) - S_2(0)\Phi(d_2)$, where

$$d_1 = \left(\ln\left(\frac{S_1(0)}{S_2(0)}\right) + \frac{\sigma_{new}^2}{2}T\right) \cdot \frac{1}{\sigma_{new}\sqrt{T}} = \frac{2\sigma^2 T}{2\sqrt{2}\sigma\sqrt{T}} = \sigma\sqrt{\frac{T}{2}}$$

$$d_2 = d_1 - \sigma_{new}\sqrt{T} = -\sigma\sqrt{\frac{T}{2}}$$

Since $d_1 = -d_2$:

$$V(0) = S_2(0) + S_1(0)\Phi(d_1) - S_2(0)\Phi(d_2) = S_1(0)\Phi(d_1) + S_2(0)(1 - \Phi(d_2)) =$$
$$= S_1(0)\Phi(d_1) + S_1(0)\Phi(-d_2) = S_1(0)\Phi(d_1) + S_2(0)\Phi(d_1)$$

$$V(0) = 200\Phi\left(\sigma\sqrt{\frac{T}{2}}\right)$$

Part b)

Question. *Derive the PDE for this contract.*

$$\mathrm{d}S_1 = rS_1\mathrm{d}t + \sigma S_1\mathrm{d}W_1$$
$$\mathrm{d}S_2 = rS_2\mathrm{d}t + \sigma S_2\mathrm{d}W_2$$

r is the risk-free interest rate, W_1 and W_2 are independent Brownian motions, meaning $\mathrm{d}W_1\mathrm{d}W_2 = 0$. Let $f(S_1, S_2, t) = Z_t$. Then by the two dimensional Itô formula:

$$\mathrm{d}Z_t = \left(\frac{\partial Z_t}{\partial t} + \frac{\partial Z_t}{\partial S_1}rS_1 + \frac{\partial Z_t}{\partial S_2}rS_2 + \frac{1}{2}\frac{\partial^2 Z_t}{\partial S_1^2}\sigma^2 S_1^2 + \frac{1}{2}\frac{\partial^2 Z_t}{\partial S_2^2}\sigma^2 S_2^2\right)\mathrm{d}t + \\ + \frac{\partial Z_t}{\partial S_1}\sigma S_1\mathrm{d}W_1 + \frac{\partial Z_t}{\partial S_2}\sigma S_2\mathrm{d}W_2$$

Based on the semester's 3rd presentation we get the riskless portfolio Π by:

$$\Pi = -Z_t + \frac{\partial Z_t}{\partial S_1}S_1 + \frac{\partial Z_t}{\partial S_2}S_2$$

$$\mathrm{d}\Pi = -\mathrm{d}Z_t + \frac{\partial Z_t}{\partial S_1}\mathrm{d}S_1 + \frac{\partial Z_t}{\partial S_2}\mathrm{d}S_2 = -\frac{\partial Z_t}{\partial t}\mathrm{d}t - \frac{1}{2}\frac{\partial^2 Z_t}{\partial S_1^2}\sigma^2 S_1^2\mathrm{d}t - \frac{1}{2}\frac{\partial^2 Z_t}{\partial S_2^2}\sigma^2 S_2^2\mathrm{d}t$$

The portfolio is riskless, so $\mathrm{d}\Pi = r\Pi\mathrm{d}t$.

$$\mathrm{d}\Pi = -rZ_t\mathrm{d}t + \frac{\partial Z_t}{\partial S_1}rS_1\mathrm{d}t + \frac{\partial Z_t}{\partial S_2}rS_2\mathrm{d}t$$

Dividing by $\mathrm{d}t$ and reordering the terms yields the final PDE:

$$\frac{\partial Z_t}{\partial t} + \frac{\partial Z_t}{\partial S_1}rS_1 + \frac{\partial Z_t}{\partial S_2}rS_2 + \frac{1}{2}\frac{\partial^2 Z_t}{\partial S_1^2}\sigma^2 S_1^2 + \frac{1}{2}\frac{\partial^2 Z_t}{\partial S_2^2}\sigma^2 S_2^2 - rZ_t = 0$$

The terminal condition is $f(S_1, S_2, T) = \max\{S_1(T), S_2(T)\}$.

Part c)

Question. *Solve the resultant PDE using Finite Difference Method.*

In order to be able to use the codes from the class (or at least something similar), we have to transform our 2D PDE into a 1D PDE. First notice that for $\lambda > 0$: $\max\{\lambda S_1, \lambda S_2\} =$

$\lambda \max\{S_1, S_2\}$. Let $x = \frac{S_1}{S_2}$ and $f(S_1, S_2, t) = S_2 u(x,t)$. Then the terminal condition is:

$$f(S_1, S_2, T) = \max\{S_1(T), S_2(T)\} = S_2(T)\max\{x, 1\}; \quad u(x,T) = \max\{x, 1\}$$

Now we have to match every term from our 2D PDE:

$$\begin{aligned}
&\frac{\partial Z_t}{\partial t} = S_2 \frac{\partial u(x,t)}{\partial t} = S_2 u_t; \quad \frac{\partial x}{\partial S_1} = \frac{1}{S_2}; \quad \frac{\partial x}{\partial S_2} = -\frac{S_1}{S_2^2} = -\frac{x}{S_2} \\
&\frac{\partial Z_t}{\partial S_1} = S_2 \frac{\partial u(x,t)}{\partial S_1} = S_2 \frac{\partial u(x,t)}{\partial x}\frac{\partial x}{\partial S_1} = S_2 u_x \frac{1}{S_2} = u_x \\
&\frac{\partial^2 Z_t}{\partial S_1^2} = \frac{\partial u_x}{\partial S_1} = \frac{\partial u_x}{\partial x}\frac{\partial x}{\partial S_1} = \frac{u_{xx}}{S_2} \\
&\frac{\partial Z_t}{\partial S_2} = S_2 \frac{\partial u(x,t)}{\partial S_2} + u(x,t)\frac{\partial S_2}{\partial S_2} = S_2 \frac{\partial u}{\partial x}\frac{\partial x}{\partial S_2} + u = S_2 u_x \left(-\frac{x}{S_2}\right) + u = u - x u_x \\
&\frac{\partial^2 Z_t}{\partial S_2^2} = \frac{\partial (u - x u_x)}{\partial S_2} = -\frac{x}{S_2}u_x + \frac{x}{S_2}u_x + \frac{x^2}{S_2}u_{xx} = \frac{x^2}{S_2}u_{xx}
\end{aligned}$$

In the last one we used that

$$\begin{aligned}
\frac{\partial u}{\partial S_2} &= \frac{\partial u}{\partial x}\frac{\partial x}{\partial S_2} = -\frac{x}{S_2}u_x \\
\frac{\partial}{\partial S_2}(-x u_x) &= -\left(\frac{\partial x}{\partial S_2}u_x + \frac{\partial u_x}{\partial S_2}x\right) = -\left(-\frac{x}{S_2}u_x + x\left(\frac{\partial u_x}{\partial x}\frac{\partial x}{\partial S_2}\right)\right) = \frac{x}{S_2}u_x + \frac{x^2}{S_2}u_{xx}
\end{aligned}$$

Now substitute everything in the 2D PDE and use that $S_1 = x S_2$ ($\Rightarrow S_1^2 = x^2 S_2^2$):

$$0 = S_2 u_t + \frac{1}{2}\sigma^2 S_1^2 \frac{u_{xx}}{S_2} + \frac{1}{2}\sigma^2 S_2^2 \frac{x^2}{S_2}u_{xx} + r S_1 u_x + r S_2 (u - x u_x) - r S_2 u$$

After careful calculations we see that r disappeared:

$$0 = S_2 u_t + \sigma^2 S_2 x^2 u_{xx}$$

Divide by $S_2 \neq 0$, and the result is

$$u_t + \sigma^2 x^2 u_{xx} = 0$$

But with that we are not ready yet, because in class the codes were written for a new variable, namely $\ln\left(\frac{S_1}{S_2}\right)$, which I will call now y. The final goal is to transform the PDE into the following form:

$$\frac{\partial V(x,t)}{\partial t} = \left[-r + \left(r - \frac{\sigma_c^2}{2}\right)\frac{\partial}{\partial x} + \frac{\sigma_c^2}{2}\frac{\partial^2}{\partial x^2}\right] V(x,t)$$

So let $y = \ln\left(\frac{S_1}{S_2}\right) = \ln(x)$, $x = e^y$. Let $u(e^y, t) = w(y, t)$. Some derivations:

$$\begin{aligned} u_x &= \frac{\partial u}{\partial x} = \frac{\partial u}{\partial y}\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{w_y}{x} \\ u_{xx} &= \frac{\partial}{\partial x}\left(\frac{w_y}{x}\right) = \left(\frac{\mathrm{d}}{\mathrm{d}x}\frac{1}{x}\right) w_y + \frac{1}{x}\left(\frac{\partial}{\partial x} w_y\right) - \frac{w_y}{x^2} + \frac{1}{x^2} w_{yy} \\ x^2 u_{xx} &= w_{yy} - w_y \end{aligned}$$

The PDE becomes

$$-w_t = \sigma^2 \frac{\partial^2 w}{\partial y^2} - \sigma^2 \frac{\partial w}{\partial y}$$

To match the required form, set $r = 0$ and $\sigma_c = \sqrt{2}\sigma$. $V(x, t) = w(y, t)$, $x = y$. Finally the boundary and terminal conditions: at maturity $\max\{x, 1\} = \max\{e^y, 1\}$. As $y_{min} \to -\infty$, $w(y_{min}, t) = \max\{0, 1\} = 1$. Otherwise $w(y_{max}, t) = \max\{e^{y_{max}}, 1\} = e^{y_{max}}$.

Part d)

I could generate two independent processes and take their maximum, and then average it out, but I found a nice theorem which I would like to present here: the Feynman-Kac theorem. I won't list all the details, but the important parts are the following: consider the following PDE:

$$\frac{\partial}{\partial t} u(x, t) + \mu(x, t)\frac{\partial}{\partial x} u(x, t) + \frac{1}{2}\sigma_k^2(x, t)\frac{\partial^2}{\partial x^2} u(x, t) - V(x, t)u(x, t) + f(x, t) = 0$$

It is defined for all $x \in \mathbb{R}$ and $t \in [0, T]$, $u(x, t)$ is the only unknown function and the terminal condition is $u(x, T) = \Psi(x)$. Then

$$u(x, t) = \mathbb{E}\left[e^{-\int_t^T V(X_s, s)\mathrm{d}s}\Psi(X_T) + \int_t^T e^{-\int_t^\tau V(X_s, s)\mathrm{d}s} f(X_\tau, \tau)\mathrm{d}\tau \middle| X_t = x\right]$$

Here X_t is an Itô process satisfying the SDE

$$\mathrm{d}X_t = \mu(X_t, t)\mathrm{d}t + \sigma_k(X_t, t)\mathrm{d}W_t,$$

where W_t is a Brownian motion. Consider our 1D PDE $u_t + \sigma^2 x^2 u_{xx} = 0$. If we match the terms with the Feynman-Kac PDE, everything is 0 except $\sigma_k(x, t) = \sqrt{2}\sigma x$ and $\Psi(x) = \max\{x, 1\}$.
If we substitute everything in the expectation formula, we get that

$$u(x_0, 0) = \mathbb{E}[\max\{X_T, 1\} | X_0 = x_0]; \quad x_0 = \frac{S_1(0)}{S_2(0)} = 1, \text{ so } u(1, 0) = \mathbb{E}[\max\{X_t, 1\}.$$

The SDE associated with our 1D PDE will be $\mathrm{d}X_t = \sqrt{2}\sigma X_t \mathrm{d}W_t$. It is necessary for us, since we will simulate the solution of this many times. By applying the Itô lemma, the solution is

$$X_t = X_0 \exp(\sqrt{2}\sigma W_t - \sigma^2 t)$$

W_t is a Brownian motion, so $W_0 = 0$ almost surely and $W_{t+l} - W_t \sim \mathcal{N}(0, l) \Rightarrow W_T - W_0 \sim \mathcal{N}(0, T) = \sqrt{T}\mathcal{N}(0, 1) = \sqrt{T}\mathcal{Z}$.
Do not forget that $f(S_1, S_2, t) = S_2 u(x, t)$, so $f(100, 100, 0) = 100\mathbb{E}[\max\{X_T, 1\}]$.
The Monte Carlo Method will be that we generate $M/2$ independent $\mathcal{Z}_i$, then also take their opposite, that way we have M standard normal samples. From them simulate $X_T^{(i)}$, $P_i = \max\{X_T^{(i)}, 1\}$, the estimation is:

$$V(0) \approx V_{MC} = \frac{100}{M} \sum_{i=1}^{M} P_i$$

By the strong law of large numbers (SLLN), as $M \to \infty$, $V_{MC} \to \mathbb{E}[\max\{X_T, 1\}]$ almost surely.

Sources

- Slides from the 3rd and 7th class
- Margrabe-formula: `https://en.wikipedia.org/wiki/Margrabe%27s_formula`
- Itô lemma: `https://en.wikipedia.org/wiki/It%C3%B4%27s_lemma`
- Feynman-Kac formula: `https://en.wikipedia.org/wiki/Feynman%E2%80%93Kac_formula`
- ChatGPT: `https://chatgpt.com/share/6af37926-03d0-83eb-82b5-195cdda340f8`
- Gemini: `https://share.gemini.google/cy954TI5Nj2U`